PHYS4071

Astrophysics & Cosmology

Course Notes

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chapterIntroduction: How to Use This Course Guide

Throughout this course guide, each topic is presented using a consistent **four-component structure** designed to support multiple learning styles:

Physics Intuition

Purpose: Builds conceptual understanding through plain language explanations.

This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept.

Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.

Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Extra

Purpose: Provides supplementary material beyond the core curriculum.

This section offers *interesting asides*, *historical notes*, *tangential topics*, citations to deeper references, or connections to cutting-edge research. Optional but enriching.

Use this when: You're curious to explore further, or when a topic connects to your research interests.

Suggested Reading Paths:

1. **Exam Preparation:** Start with *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Extra* → *Theory* → *Question Examples*
3. **Quick Review:** *Theory* → *Question Examples*

chapterThe Expanding Universe
section[Hubble's Discovery]Hubble's Discovery
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna.

subsection[Theory]Theory

Theory

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus.
The **Hubble–Lemaître Law** relates the recession velocity of a galaxy to its proper distance d:

$$v = H_0 d, \tag{1.1}$$

where $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the Hubble constant (Planck 2018).
The cosmological redshift is defined via the scale factor $a(t)$:

$$1 + z = \frac{a(t_0)}{a(t_e)} = \frac{1}{a(t_e)}, \tag{1.2}$$

using the convention $a(t_0) = 1$ today.
Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis.

subsection[Question Examples]Question Examples

Question Example 1.1: Recession Velocity from Redshift

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero.

Solution. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante.

subsection[Extra]Extra

Extra

Quisque ullamcorper placerat ipsum. Cras nibh.

section[The Cosmological Redshift]The Cosmological Redshift
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl.

subsection[Theory]Theory

Theory

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna.

subsection[Question Examples]Question Examples

Question Example 1.2: Computing Cosmic Distance

Sed commodo posuere pede. Mauris ut est.

Solution. Ut quis purus. Sed ac odio.

subsection[Extra]Extra

Extra

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo.

chapterFriedmann Equations & Cosmological Dynamics
section[The FLRW Metric]The FLRW Metric
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante.

subsection[Theory]Theory

Theory

The **FLRW metric** describes a homogeneous, isotropic universe:

$$ds^2 = -c^2\,dt^2 + a(t)^2\left[\frac{dr^2}{1-kr^2} + r^2\,d\theta^2 + r^2\sin^2\theta\,d\phi^2\right], \tag{2.1}$$

with curvature parameter $k \in \{-1, 0, +1\}$.
Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui.

subsection[Question Examples]Question Examples

Question Example 2.1: Proper Distance in FLRW

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

Solution. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien.

subsection[Extra]Extra

Extra

Etiam euismod. Fusce facilisis lacinia dui.

section[The Friedmann Equations]The Friedmann Equations
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor.

subsection[Theory]Theory

Theory

Inserting the FLRW metric into the Einstein field equations gives the **Friedmann equations**:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{2.2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{2.3}$$

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim.

subsection[Question Examples]Question Examples

Question Example 2.2: Scale Factor of a Flat Matter-Dominated Universe

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi.

Solution. For $k = 0$, $\Lambda = 0$ and pressureless dust ($p = 0$), Eq. (2.2) reduces to $\dot{a}^2 \propto a^{-1}$, giving:

$$a(t) \propto t^{2/3}. \tag{2.4}$$

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo.

subsection[Extra]Extra

Extra

Nulla mattis luctus nulla. Duis commodo velit at leo.

chapterEnergy Budget in the ΛCDM Model
section[Density Parameters & Critical Density]Density Parameters & Critical Density
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio.

subsection[Theory]Theory

Theory

Define the **critical density** and dimensionless density parameters:

$$\rho_{\text{crit}}(z) = \frac{3H(z)^2}{8\pi G}, \qquad \Omega_i(z) = \frac{\rho_i(z)}{\rho_{\text{crit}}(z)}. \tag{3.1}$$

Each component scales with redshift according to its equation of state w_i:

$$\rho_i(z) = \rho_{i,0}(1+z)^{3(1+w_i)}. \tag{3.2}$$

Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh.

subsection[Question Examples]Question Examples

Question Example 3.1: Computing Critical Density Today

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra.

Solution. Nullam varius. Etiam dignissim elementum metus.

subsection[Extra]Extra

Extra

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus.

section[The Flat ΛCDM Model]The Flat ΛCDM Model
subsection[Physics Intuition]Physics Intuition

Physics Intuition

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus.

subsection[Theory]Theory

Theory

In the **flat ΛCDM** model ($\Omega_k = 0$, $\Omega_{\mathrm{m},0} + \Omega_\Lambda + \Omega_{\mathrm{r},0} = 1$):

$$\frac{H(z)^2}{H_0^2} = \Omega_{\mathrm{m},0}(1+z)^3 + \Omega_{\mathrm{r},0}(1+z)^4 + \Omega_\Lambda. \tag{3.3}$$

Current Planck 2018 best-fit values:

$$\Omega_{\mathrm{m},0} \approx 0.315, \quad \Omega_\Lambda \approx 0.685, \quad \Omega_{\mathrm{r},0} \approx 9.2 \times 10^{-5}.$$

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci. Pellentesque mattis lacinia elit.

subsection[Question Examples]Question Examples

Question Example 3.2: Age of the Universe in Flat ΛCDM

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus.

Solution. The lookback-time integral is:

$$t_0 = \int_0^\infty \frac{dz}{(1+z)\,H(z)}, \tag{3.4}$$

with $H(z)$ from Eq. (3.3). Morbi purus. Nulla a est sit amet purus venenatis iaculis.

subsection[Extra]Extra

Extra

Maecenas non massa. Vestibulum pharetra nulla at lorem.

SECTION[ENERGY BUDGET EVOLUTION]ENERGY BUDGET EVOLUTION
SUBSECTION[VISUALIZING THE ENERGY DENSITY]VISUALIZING THE ENERGY DENSITY

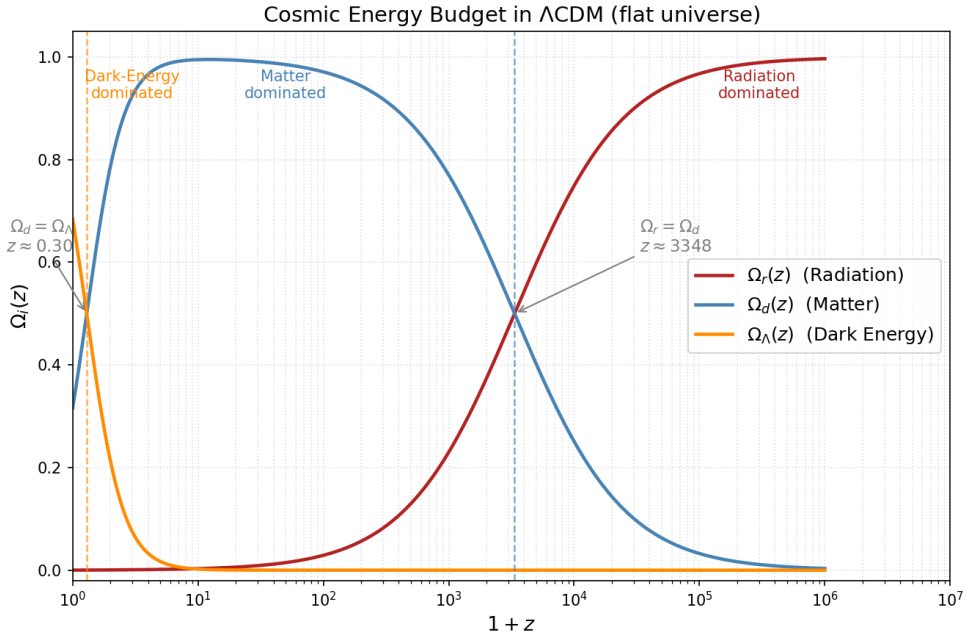

Figure 3.1: **Evolution of energy density components in the flat ΛCDM model.** The figure shows how the fractional energy density (normalized to critical density) of matter (dark and baryonic), radiation, and dark energy evolve with redshift z. At early times ($z \gg 1$), radiation dominated; at intermediate times, matter dominated; and at late times ($z \lesssim 1$), dark energy becomes dominant. This transition explains the accelerated expansion observed in the present-day universe.

EXTRA

THE VISUALIZATION OF ENERGY BUDGET COMPONENTS PROVIDES CRUCIAL INSIGHTS INTO COSMIC EVOLUTION. EARLY UNIVERSE PHYSICS WAS RADIATION-DOMINATED, WHILE THE MATTER-DOMINATED ERA SHAPED STRUCTURE FORMATION AND GALAXY EVOLUTION. THE RECENT TRANSITION TO DARK-ENERGY DOMINATION (AROUND $z \sim 0.7$) EXPLAINS THE OBSERVED COSMIC ACCELERATION DETECTED THROUGH TYPE IA SUPERNOVAE OBSERVATIONS CARMELI$_2$001.

chapterMathematical Tools & Derivations
section[Useful Identities]Useful Identities
Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum. Suspendisse condimentum, tortor at egestas posuere, neque metus tempor orci, et tincidunt urna nunc a purus. Sed facilisis blandit tellus. Nunc risus sem, suscipit nec, eleifend quis, cursus quis, libero. Curabitur et dolor. Sed vitae sem.
section[Differential Geometry Primer]Differential Geometry Primer
Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim. Aliquam erat volutpat. Proin sit amet nulla ut eros consectetuer cursus. Phasellus dapibus aliquam justo. Nunc laoreet. Donec consequat placerat magna.

chapterDeclaration of AI Assistance
section[Use of Artificial Intelligence in This Document]Use of Artificial Intelligence in This Document
This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes, and designing the pedagogical framework (Physics Intuition → Theory → Question Examples → Extra).
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with cosmology conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

Compiled by: automatoc-ti | *Date: Saturday 7th March, 2026*

chapterConstants & Conversion Factors
section[Physical Constants]Physical Constants

Quantity	**Symbol**	**Value (SI)**
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$

section[Conversion Factors]Conversion Factors
Donec vel nibh ut felis consectetuer laoreet. Donec pede. Sed id quam id wisi laoreet suscipit. Nulla lectus dolor, aliquam ac, fringilla eget, mollis ut, orci.

REFERENCES